

Question	Marking Guidance	Mark	Comments
08.1	- Oxygen produced in light-dependent reaction; - The faster (oxygen) is produced, the faster the light-dependent reaction;	2	
08.2	35–36 μmol oxygen per mg chlorophyll;	2	Correct difference at $500 \mu\text{mol photons m}^{-2} \text{s}^{-1}$ or incorrect difference but division by 4 shown = 1 mark
08.3	At all light intensities, chloroplasts from mutant plants: - Have faster production of ATP and reduced NADP; (So) have faster/more light-independent reaction; (So) produce more sugars that can be used in respiration; (So) have more energy for growth; - Have faster/more synthesis of new organic materials;	4 max	Accept converse points if clear answer relates to non-mutant plants
